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Course - Introduction to Problem Solving and Programming

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1) INTRODUCTION

Managing personal finance is a significant concern in daily life. Many people cannot remember where their money went, how much they have left at the end of the month, and so on. The project discussed here focuses on the implementation of a simple Personal Finance Manager using Python. The goal of this project was to devise a small tool for users to add income and expenses and then view the summary of their financial activity.

The project focuses on the basics of Python: modular programming, functions, file handling, and the creation of a menu-driven console interface. This is intentionally kept simple so the logic remains easy to understand and implement.

2) PROBLEM STATEMENT

Most people track their spending either very informally or not at all, which leads to a lot of confusion and obscurity regarding their financial condition. It is difficult to maintain daily records of income and expenses without an appropriate tool.

This project solves this problem by offering a simple Python-based solution, which uses a file-based system that stores the transactions and then automatically generates summary reports.

3) FUNCTIONAL REQUIREMENTS

- User can add either income or expense
- User inputs amount and description
- All transactions are recorded inside a text file
- Persistent storage
- Shows total income
- Shows total expenses
- Display final balance
- Menu-Based Navigation

4) NON-FUNCTIONAL REQUIREMENTS

- Easy to use for beginners
- Simple text-based interface
- Data is stored in files to avoid loss
- Program handles invalid inputs safely
- Modular structure improves readability and future updates
- Fast execution since only basic Python is used
- Input validation for incorrect data

5) SYSTEM ARCHITECTURE

This project uses a modular architecture where each major function is placed in a separate Python file:

- `main.py` → Controls program flow
- `transaction_manager.py` → Adds new transactions
- `file_manager.py` → Reads and writes to file
- `report_manager.py` → Generates financial summary

6) SCREENSHOTS

```
==== Personal Finance Manager ====
1. Add Income
2. Add Expense
3. View Transactions
4. Delete Transaction
5. Show Summary Report
6. Show Monthly Report
7. Save & Exit
Enter choice: 2
Enter expense amount: 200
Enter expense category (e.g., food, travel): food
Enter note (optional): cafe
Expense added (id=2).
```

```
==== Personal Finance Manager ====
1. Add Income
2. Add Expense
3. View Transactions
4. Delete Transaction
5. Show Summary Report
6. Show Monthly Report
7. Save & Exit
Enter choice: 1
Enter income amount: 2000
Enter income source/category (optional): pocket money
Income added (id=1).
```

```
1  id,date,type,amount,category,note
2  1,2025-11-24,income,2000.00,pocket money,
3  2,2025-11-24,expense,200.00,food,cafe
4
```

7) TESTING APPROACH

- **Valid Inputs**
 - Normal income and expense entries
- **Invalid Inputs**
 - Letters instead of numbers
 - Empty description
- **File Handling**
 - Verify file updates correctly
- **Report Accuracy**
 - Totals and balance correct
- **Menu Flow**
 - Works without crashes

8) CHALLENGES FACED

- Deciding how to structure the project into modules
- Handling wrong inputs without breaking the program
- Ensuring the file format stays clean
- Making the report easy to read
- Solving the errors faced in creating the code

9) LEARNING AND KEY TAKEAWAYS

- Better understanding of Python file handling
- Importance of modular design
- How real applications store and manage data
- Writing simple but maintainable code
- Planning workflows before coding improves clarity

10) REFERENCES

- Python Official Documentation
- Programming Textbooks
- Online Python Tutorials